Jiaheng Huang

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EDUCATION

Northeastern University — Boston, MA **Expected May 2029**
Candidate for Bachelor of Science in Mechanical Engineering
- **Relevant Coursework:** Statics, Materials Science, Differential Equations, Physics 2 (E&M), Cornerstone of Engineering 1 & 2, Intro to Program Design

Mission San Jose High School — Fremont, CA **May 2025**

TECHNICAL SKILLS

- **CAD & Design:** SolidWorks, Fusion 360, Onshape, AutoCAD
- **Programming:** Python, MATLAB, MicroPython, Excel
- **Manufacturing:** 3D printing, manual turning, waterjet cutting, laser cutting
- **Certifications:** Six Sigma Green Belt (IISE), SolidWorks CSWA (in progress)

PROJECTS

Save SpongeBob: Sustainability Museum Exhibit — Northeastern University **January 2026 – April 2026**
- Designed a portable exhibit teaching 4th–5th graders about overfishing via a magnetic fishing game, chosen from research showing hands-on play holds children's attention longest.
- Bridged mechanical-electrical integration on a team of 4, modeling the fish platform, rod assembly, and LED board in SolidWorks with tolerances set for 3D printing or laser cutting.
- Controlled a Raspberry Pi Pico from MATLAB over serial to run a load cell, buttons, LEDs, and a motorized tub.
- Delivered a folding-table-sized build for under **$100**, iterating from cardboard mockups through prototype testing.

Electric Go-Kart — Fremont, CA **May 2025 – July 2025**
- Converted a vintage race kart to electric drive, restoring the chassis and rebuilding the hydraulic brakes.
- Integrated a custom battery pack with a VESC controller, designing 3D-printed brackets and a sheet-metal motor mount.
- Tuned field-oriented control alongside chassis setup (toe, caster, tire pressure, track width) to refine torque and handling.
- Validated performance through on-track testing, achieving **4.23 s 0–45 mph acceleration** and **14.39 kW peak output**.

Robot Actuator **September 2026 – Present**
- Designing a 3D-printed quasi-direct-drive planetary actuator targeting 5 N·m continuous torque to drive a robot arm.

LEADERSHIP & ACTIVITIES

Northeastern Electric Racing — Boston, MA **January 2026 – Present**
Vehicle Dynamics Member
- Machined suspension jig parts and frame weld tabs; turned suspension bushings on a manual lathe.

NURobotics — Boston, MA **September 2026 – Present**
Combat Robotics Member, 1 lb Plastic Antweight
- Co-designing a 2WD horizontal-spinner combat robot in Onshape for 3D-printed fabrication.

Drum Corps International — Concord, CA **January 2023 – May 2024**
Mellophone Performer, Blue Devils B & Mandarins
- Performed with Blue Devils B, a top-ranked Open Class corps, sustaining 12-hour rehearsal days on the California circuit.
- Earned a contract with the Mandarins, a Top 5 World Class corps, through a two-round audition, training alongside music performance majors and professional musicians.

WORK EXPERIENCE

In-Home Supportive Services (IHSS) — Fremont, CA **November 2023 – August 2025**
Care Provider
- Provided independent in-home care over 21 months, managing daily tasks and adapting to evolving patient needs.

Friends of Children with Special Needs (FCSN) — Fremont, CA **February 2019 – June 2023**
Volunteer Teaching Assistant, LEGO Robotics
- Coached 10–15 students with special needs in LEGO Robotics with the lead instructor and 3 volunteers.
- Adapted guidance to each student's learning needs, increasing engagement and independent task completion.

INTERESTS

- Rock Climbing, Music Production, Brass Performance, Photography